DRAINAGE

Highway drainage facilities carry water across the right-of-way and remove stormwater from the roadway itself. Drainage facilities include;

- bridges,
- culverts,
- channels,
- curbs,
- gutters, and
- various types of drains.

Hydraulic capacities and locations of such structures should be designed to take into consideration;

- damage to upstream and downstream property and
- to secure as low a degree of risk of traffic interruption by flooding as is consistent with the importance of the road, the design traffic service needs.

Whenever there is a rainfall, some of the rainwater infiltrated into the ground and stored as ground water and some of the portion may evaporate into the atmosphere. Other than these losses, the water left on the surface is called as **run off**.

The method of estimating the run off is called **hydrologic analysis**. To estimate the maximum quantity of water expected to reach the drainage system is the main objective of hydrologic analysis.

The factors affecting run off and they are

- Rate of rain fall
- Moisture condition
- Soil type
- Ground cover presence
- Topography

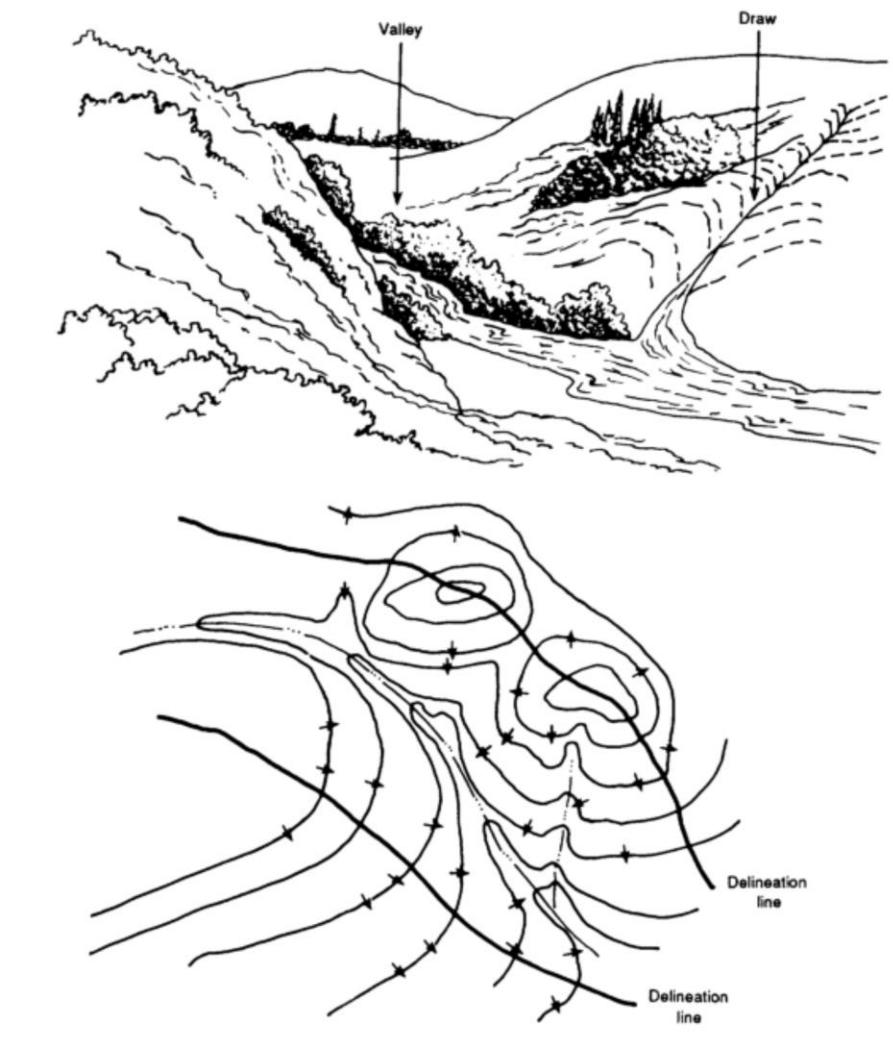
Other than the above factors, rain fall intensity, occurrence of storms in that area are to be studied from the old records. Hence, maximum run off can be estimated to build safe surface drainage system.

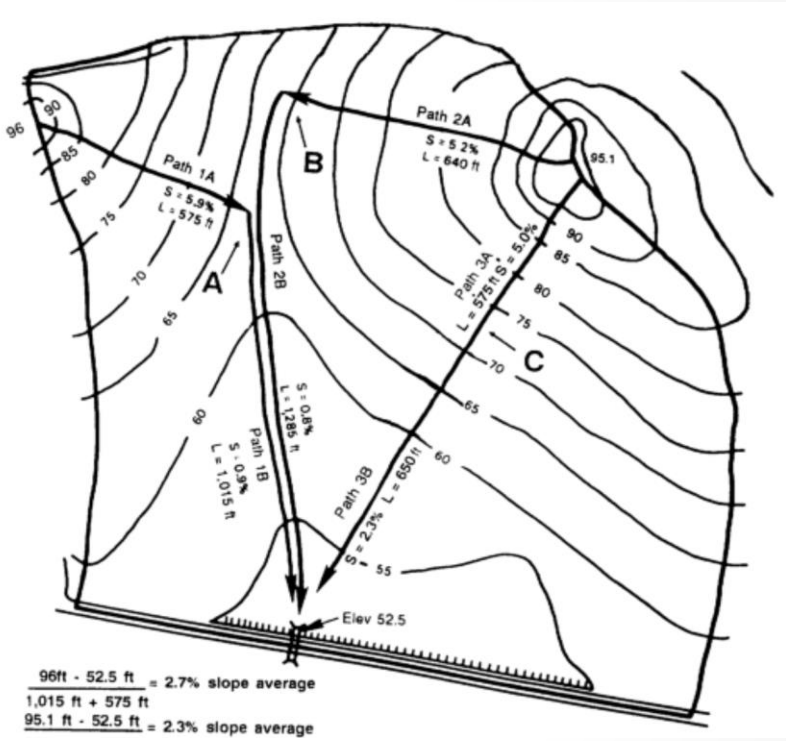
The run off can be calculated by below formula

$$\mathbf{Q = C * i * } \boldsymbol{A_d}$$

Q = run off (m^3/sec)
C = run off coefficient
i = intensity of rain fall (mm/sec)
A_d = area of drainage (m^2)

Area of drainage





The run off can be calculated by below formula

$$\mathbf{Q = C * i * A_d}$$

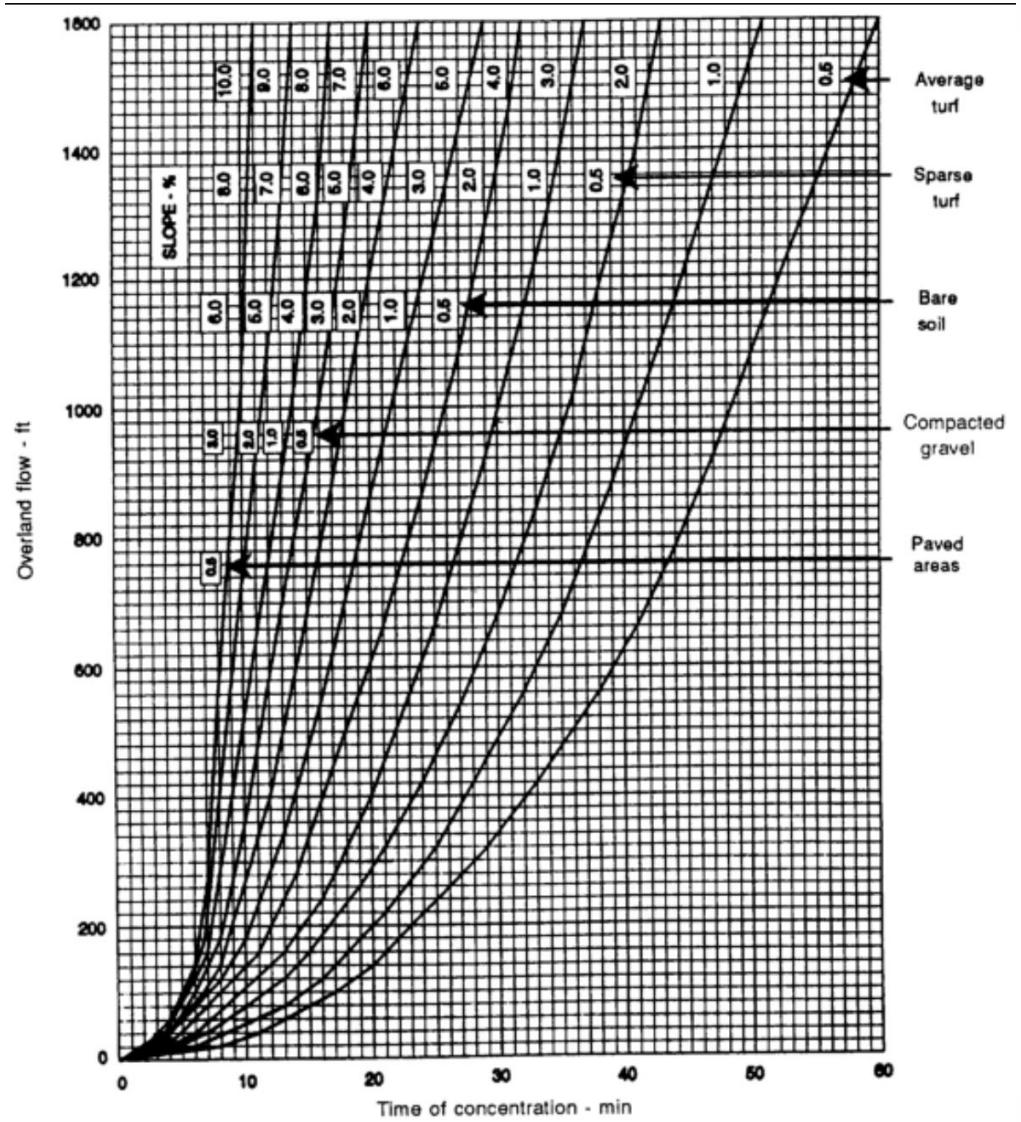
Q = run off (m^3/sec)
C = run off coefficient
i = intensity of rain fall (mm/sec)
A_d = area of drainage (m^2)

Type of Surface	Coefficient of run off
Pervious soil surface	0.05 – 0.30
Soil covered with turf	0.30 – 0.55
Impervious soil	0.40 – 0.65
Gravel & WBM roads	0.35 – 0.70
Bituminous & C.C roads	0.80 – 0.90

If the drainage area contains different surfaces in it then run off coefficient is calculated as;

$$\mathbf{C} = \frac{A_1 * C_1 + A_2 * C_2 + A_3 * C_3}{A_1 + A_2 + A_3}$$

Where C_1, C2, C3 are run off coefficients for different surfaces and A_1, A2, A3 are their respective areas.



NOTE: It is valid to interpolate between percent slope curves.

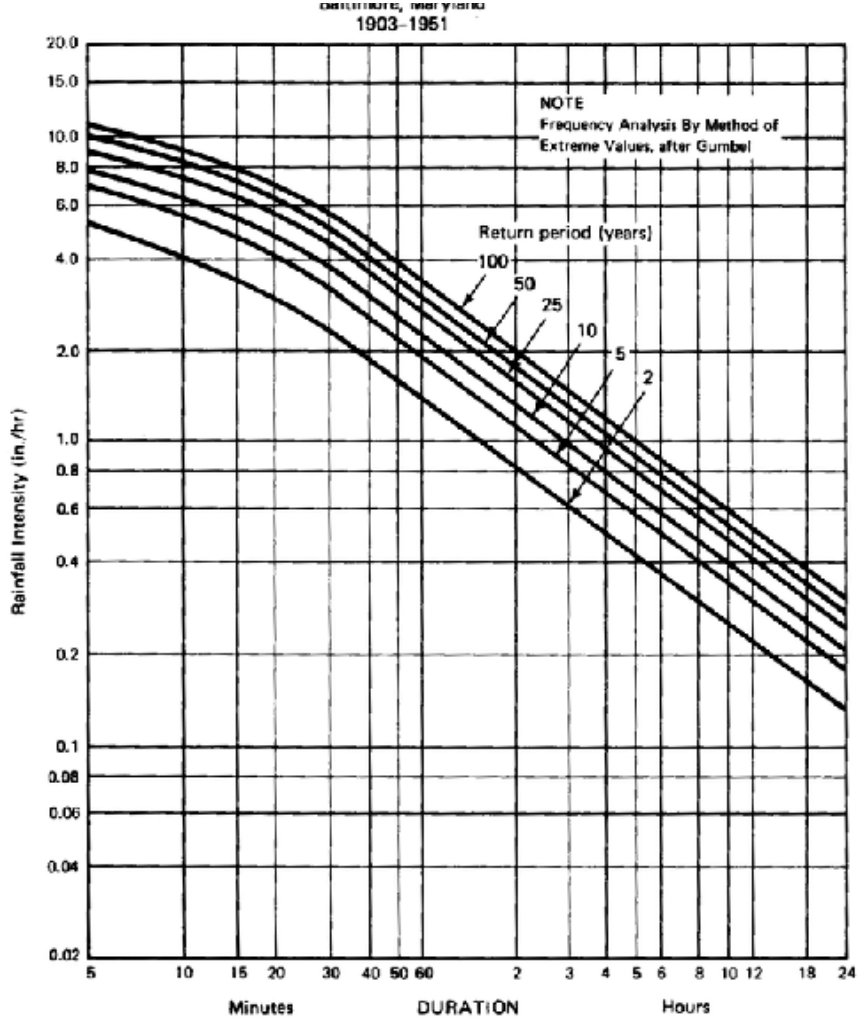


FIGURE 4–4 Rainfall intensity-duration-frequency. (National Weather Service, 1961.)

$$\mathbf{Q} = \mathbf{V} * \boldsymbol{A}$$

Q = run off (m^3/sec)
V = Allowable velocity in channel (m/sec)
A = cross section area of channel

Soil type	**Allowable velocity (m/sec)**
Sand or silt	0.30 – 0.50
Loam	0.60 – 0.90
Clay	0.90 – 1.50
Gravel	1.20 – 1.50
Soil with grass	1.50 – 1.80



Allowable Velocity

Manning's formula:

$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

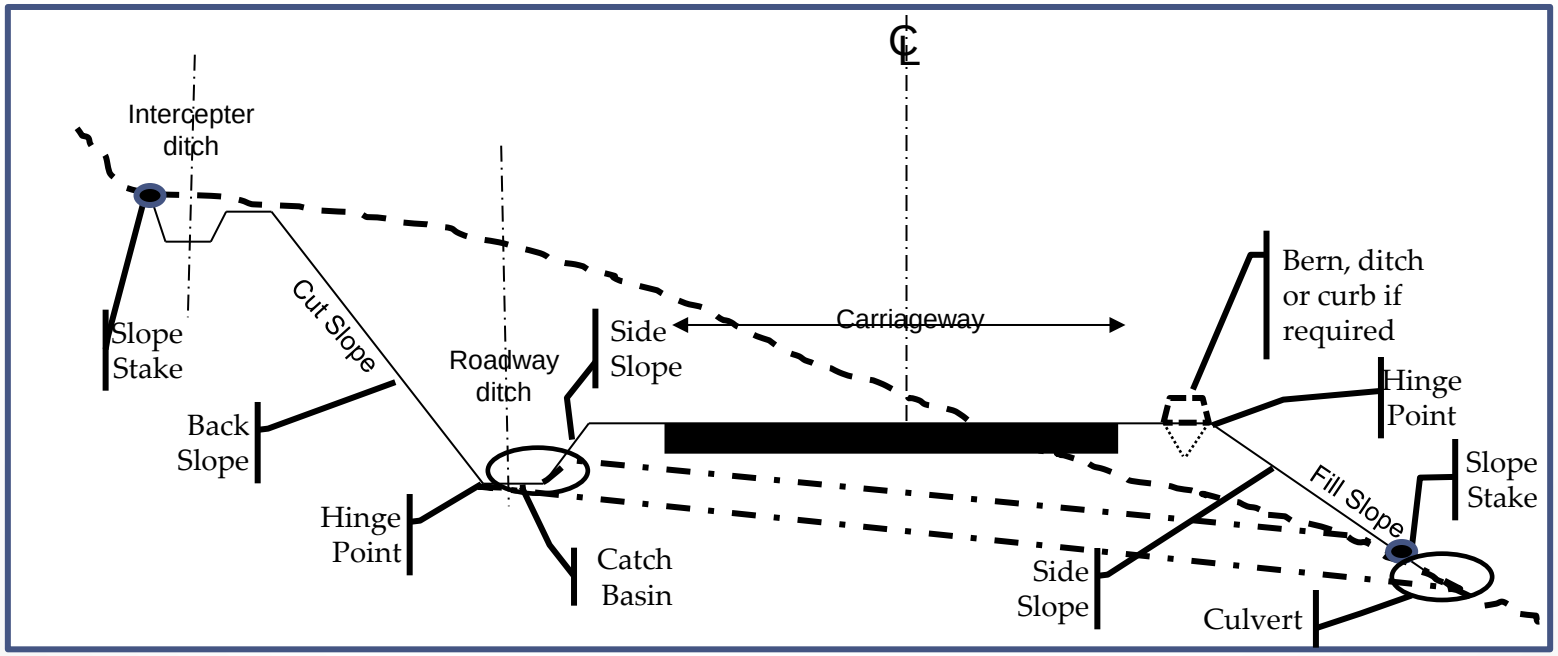
V = Allowable velocity in channel (m/sec)
n = Manning's roughness coefficient
R = Hydraulic radius (m) :the ratio of area of the channel to its wetted perimeter
S = Longitudinal Slope of Channel

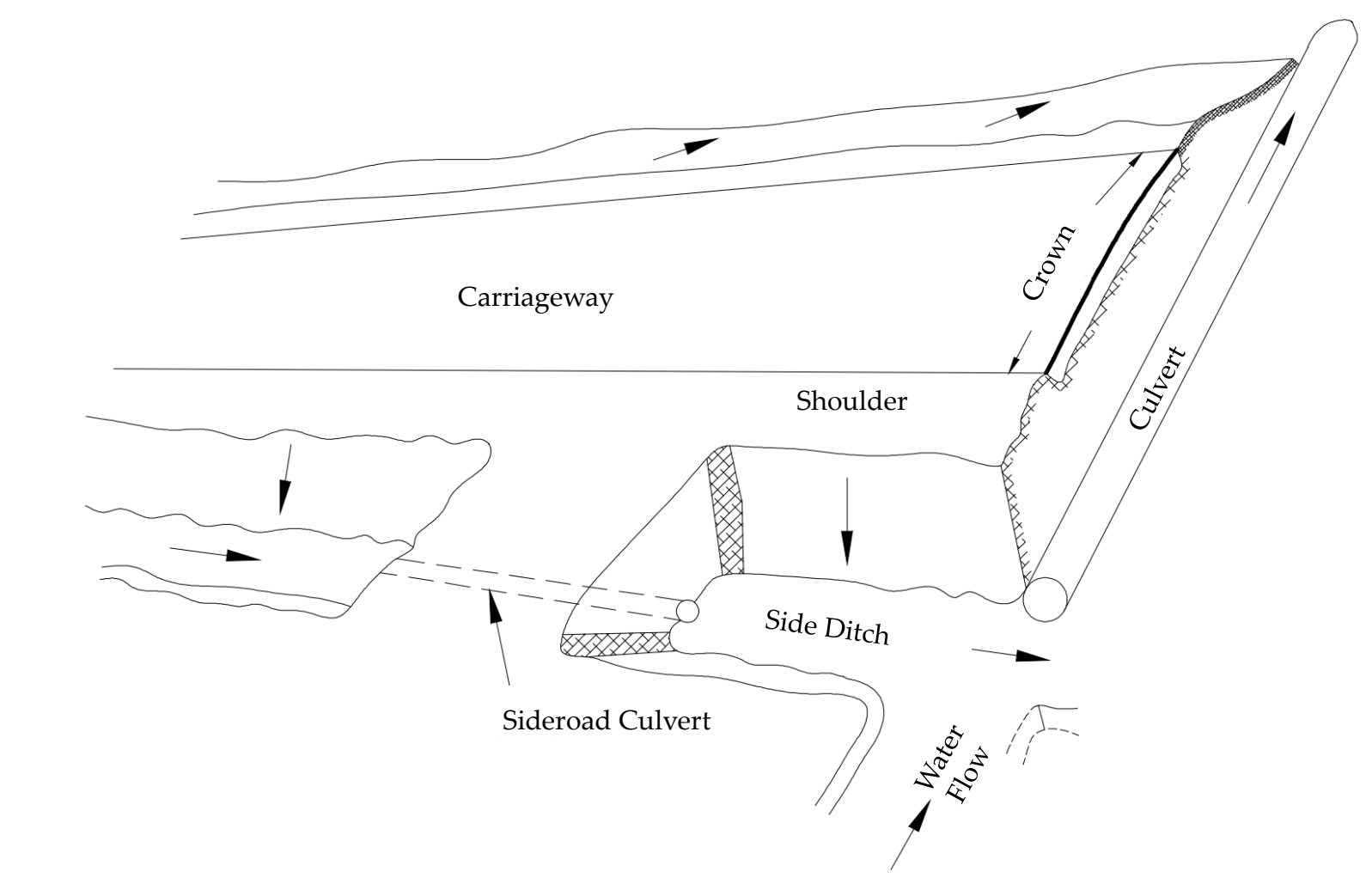
Lining material	**Manning's roughness coefficient, n**
Ordinary soil	0.02
Soil with grass layer	0.05 – 0.10
Concrete lining	0.013
Rubble lining	0.04

DRAINAGE

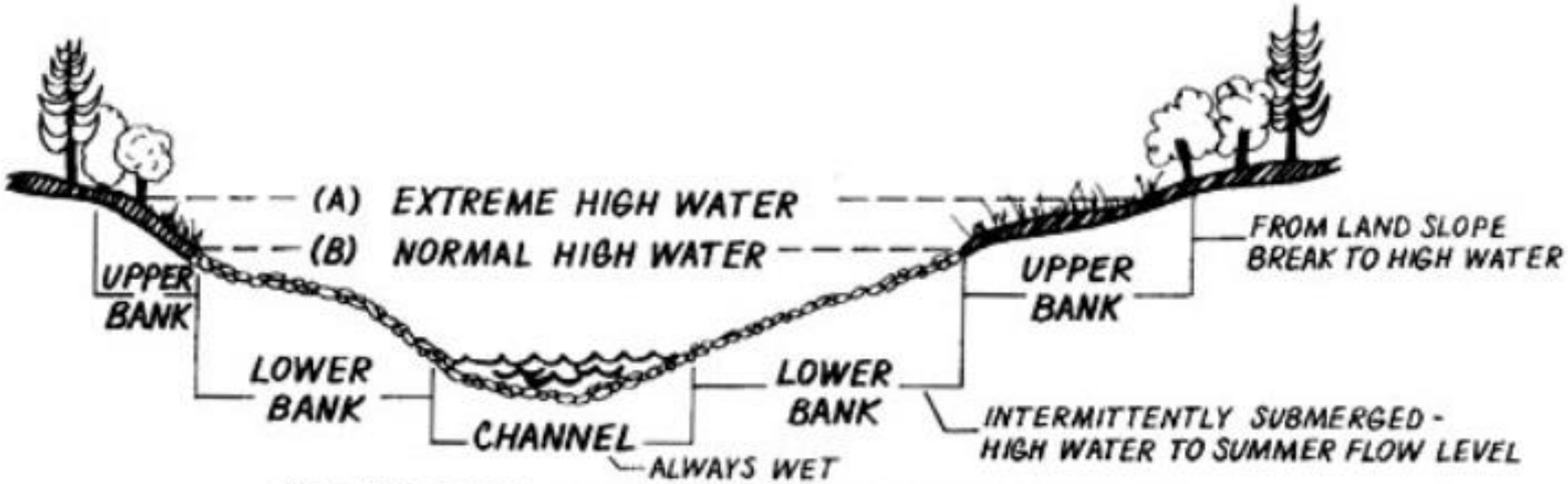
Ditch: A small artificial channel cut through earth or rock to carry water for irrigation or drainage. A long narrow cut made in the earth to bury pipeline, cable, or similar installations.
Slope, crossfall: Surface cross slopes are necessary on all components of the crosssection to facilitate drainage.

Slope stake: Slope staking is a trial and error process of finding the point where the slope intercepts the natural ground. This point is referred to as the “catch point” or “slope catch.”
Intercepting ditch: An interceptor ditch is a small ditch or channel constructed to intercept and convey water to an area where it can be safely discharged.
Side slope: A long narrow hole that is dug along a road, field, etc., and used to hold or move water.




Carriageway
Crown
Culvert
Shoulder
Side Ditch
Sideroad Culvert
Water
Flow

DRAINAGE

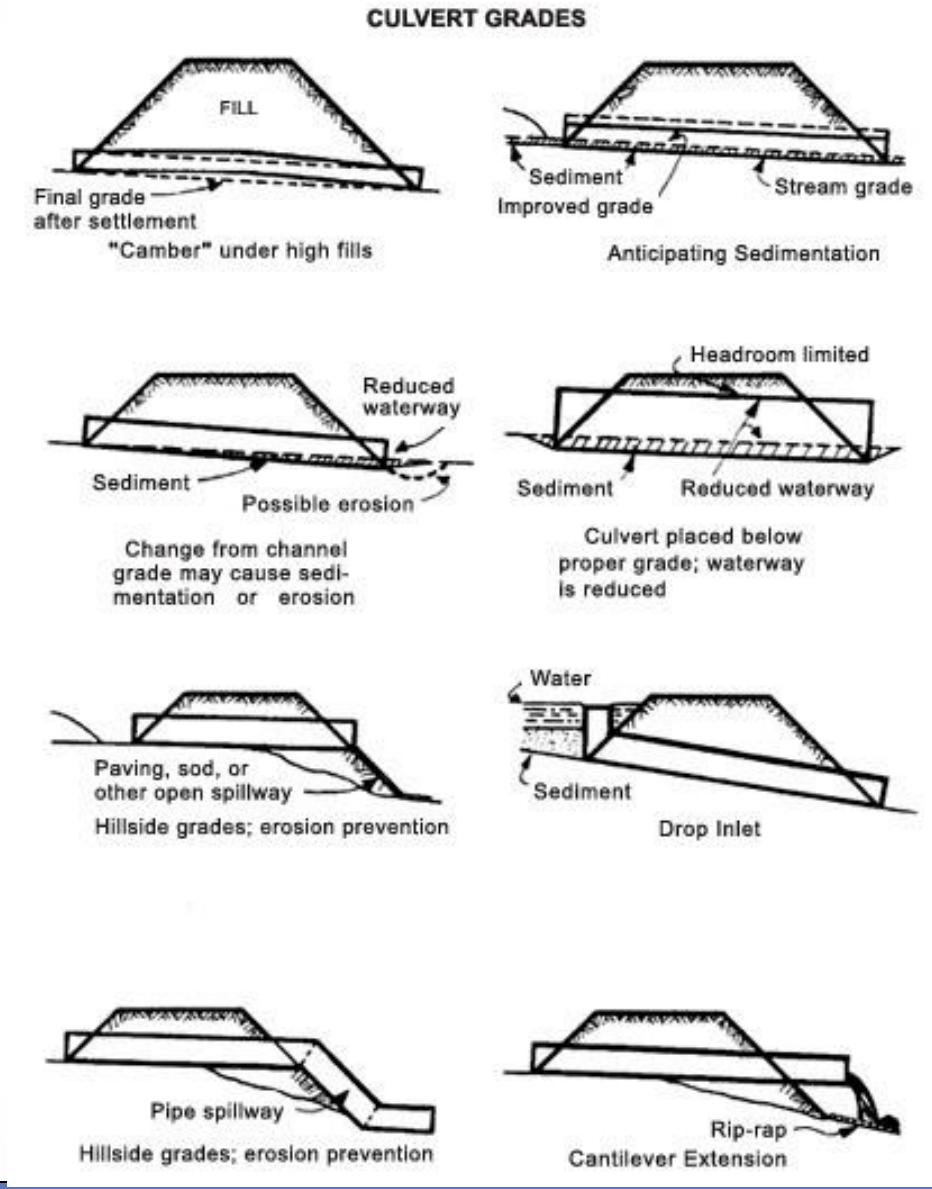


(A) FLOODING. THE FLOOD PLAIN IS COMPLETELY COVERED.

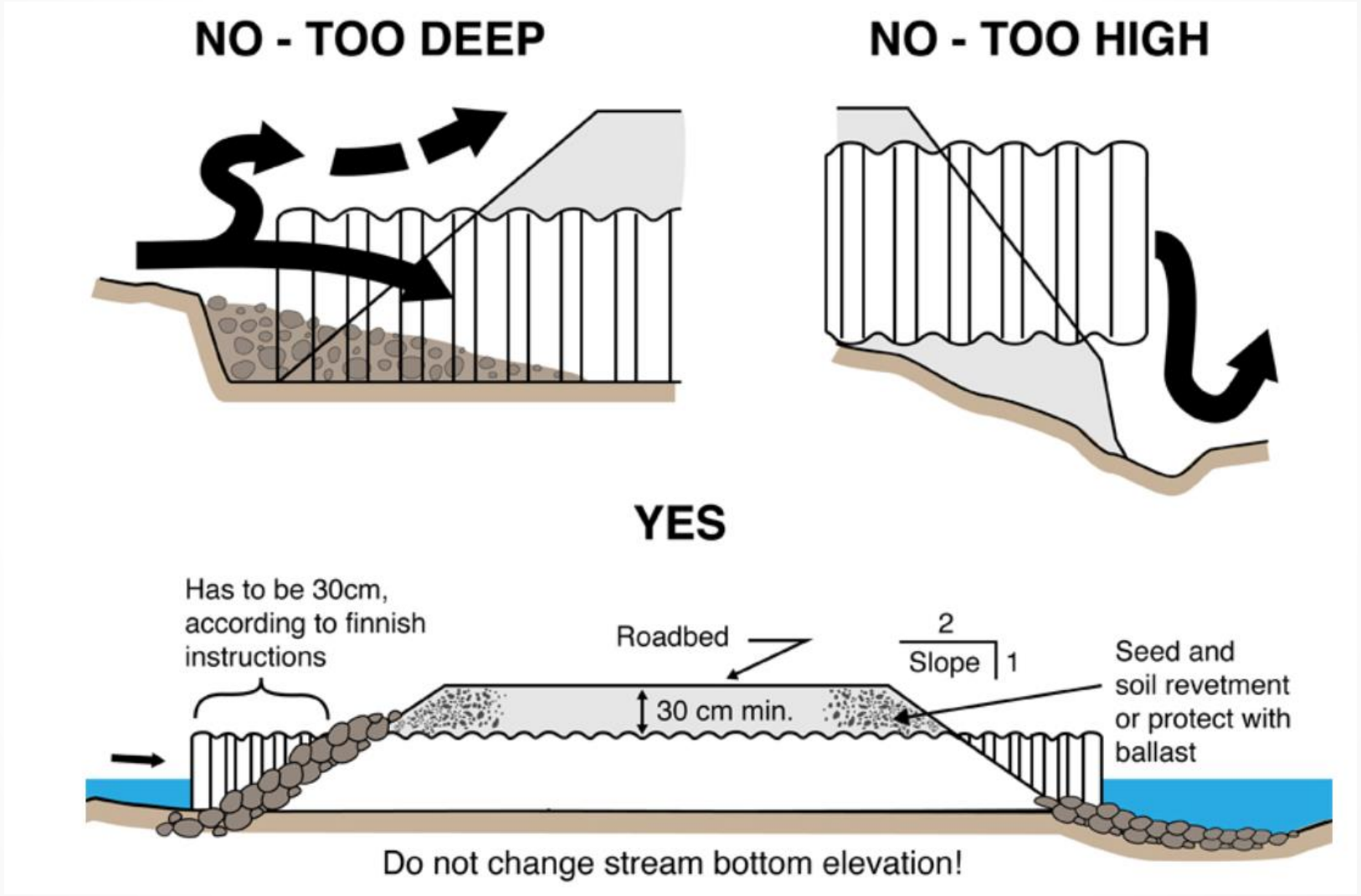
(B) HIGH WATER. THE CHANNEL FULL TO THE NORMAL HIGH WATER LINE.

STREAM CHANNEL TERMS

DRAINAGE


CULVERT GRADES
FILL
Final grade
after settlement
"Camber" under high fills
Sediment
Improved grade
Stream grade
Anticipating Sedimentation
Reduced
waterway
Sediment
Possible erosion
Change from channel
grade may cause sedi-
mentation or erosion
Headroom limited
Sediment
Reduced waterway
Culvert placed below
proper grade; waterway
is reduced
Paving, sod, or
other open spillway
Hillside grades; erosion prevention
Water
Sediment
Drop Inlet
Pipe spillway
Hillside grades; erosion prevention
Rip-rap
Cantilever Extension

DRAINAGE


NO - TOO DEEP
NO - TOO HIGH
YES
Has to be 30cm, according to finnish instructions
Roadbed
2
Slope
1
30 cm min.
Seed and soil revetment or protect with ballast
Do not change stream bottom elevation!